Air Pollution Patterns in Wisconsin

- County-level PM2.5 spatial analysis for 2024
- Legacy replication plus portfolio-grade rebuild
- Yifang Qiu contribution highlighted

Motivation and Research Questions

- Which monitored counties had the highest descriptive annual mean PM2.5?
- How sparse was Wisconsin's county monitor coverage?
- Do monitored counties show robust spatial clustering?

Data Sources and Processing

- EPA Air Trends / AirData references provide pollutant context.
- GeoData@Wisconsin supplies the archived Wisconsin county boundary layer.
- Population context uses supplied tables tied to the 2020 Census.

Data Quality and Monitor Coverage

- 72 total counties in the shapefile.
- 16 counties with PM2.5 benchmark values.
- Unmonitored counties are kept as missing in statewide mapping.

Methods

- Legacy estimator: mean daily PM2.5 by site, then unweighted mean across sites by county.
- Legacy spatial model: Queen contiguity among monitored counties only.
- Robustness model: KNN weights using projected county representative points.

Statewide Choropleth

- Statewide map preserves all counties.
- Gray fill marks no monitoring data.
- County means are descriptive monitoring summaries, not regulatory design values.

County Ranking

- Highest monitored county: Grant.
- Lowest monitored county: Monroe.
- Milwaukee and Dane each have two monitoring sites in the legacy benchmark table.

Global and Local Spatial Results

- Legacy Moran's I = -0.140.
- Legacy Queen weights contain disconnected components and island counties.
- Local Moran categories should not be described as robust clusters under the legacy graph.

Sensitivity, Limitations, and Conclusion

- KNN weights remove zero-neighbor counties.
- Sparse coverage limits statewide inference.
- The most defensible conclusion is descriptive rather than causal or regulatory.

References and Team Contributions

- EPA Air Trends and AirData
- GeoData@Wisconsin / U.S. Census TIGER/Line

- Yifang Qiu: analysis, spatial joining, choropleth mapping, county ranking, Global Moran's I, Local Moran's I, and visualization